

Mid-term Exam 1 Example Questions

Questions:

1. Calculate bulk density, porosity, gravimetric water content, and volumetric water content for a cylindrical soil sample 10.0 cm long and 7.50 cm in diameter, with a wet mass of 685 g and an oven-dry mass of 572 g. Assume a particle density of 2.65 g/cm^3 . Assume the wet and dry mass does not contain the mass of the ring.
2. What is a contact angle? What does it depend on? How is it used to determine hydrophobic and hydrophilic surfaces?
3. Define capillarity. Explain why it occurs. Describe one important effect of capillary rise.
4. Calculate the height of capillary rise for a soil with an average pore-diameter of 0.01 mm. Assume a contact angle of 20° , surface tension (γ) = $7.3 \times 10^{-2} \text{ N m}^{-1}$, density of water = 998 kg m^{-3} , and acceleration due to gravity = 9.81 m s^{-2} .
5. Calculate soil water storage for the following soil profile. The A horizon is 32.0 cm thick and has a volumetric water content of $0.18 \text{ cm}^3 \text{ cm}^{-3}$. The B horizon is 30.0 cm thick and has a volumetric water content of $0.17 \text{ cm}^3 \text{ cm}^{-3}$.
6. After 2.5 cm of rainfall infiltrates the soil profile in the previous question, what is the new volumetric water content of the A horizon? Assume no water moved from the A horizon to the B horizon.
7. Consider a 1-L graduated cylinder with an initially uniform (i.e. well-stirred) dilute suspension of soil particles in water. Samples will be withdrawn from 15. cm below the water surface for particle size analysis. Assume terminal velocity is obtained right after stirring ends. How long should we wait before withdrawing the samples to be certain that:
 - a. All the sand-sized particles have dropped below the sampling depth?
 - b. All the silt-sized particles have dropped below the sampling depth?
8. Define surface tension and describe its relationship with hydrogen bonding and capillarity.
9. Define soil water potential. Define each of its three primary components. What are its units?
10. Draw a figure showing typical soil water retention curves for a coarse-textured soil and a fine-textured soil. Draw both figures in the same chart. Label the axes. Specify the units and choose realistic maximum and minimum values for each axis. Write a short paragraph in which you define the soil water retention curve and describe the differences between the curves for coarse-textured soil versus fine-textured soil.

11. Define hysteresis as it applies to the soil water retention curve.
12. Define what is a “system at equilibrium”
13. Using the soil water retention model proposed by Campbell in 1974, calculate the plant available water capacity of a sandy loam and a silt loam soil. Assume that field capacity is reached when the soil has a matric potential of -10 J kg^{-1} and that plants cannot longer extract water when the soil matric potential reaches $-1,500 \text{ J kg}^{-1}$.
14. A loamy sand soil with a volumetric water content of $0.185 \text{ cm}^3 \text{ cm}^{-3}$ is in contact (see sketch) with a clay loam soil that has a volumetric water content of $0.210 \text{ cm}^3 \text{ cm}^{-3}$. In what direction does the water move? Assume no solutes.
 - a. Water moves from the Loamy sand to the Clay loam soil
 - b. Water moves from the Clay loam to the Loamy sand soil
 - c. There is no net movement because the soil is already in hydraulic equilibrium

Hint: The soils are side-by-side, which means that the gravitational potential is the same for both soils.

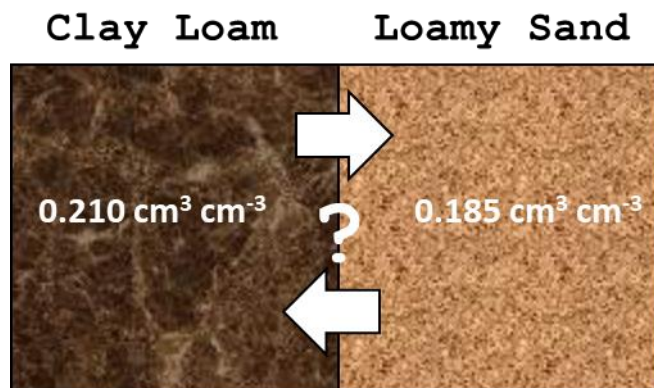
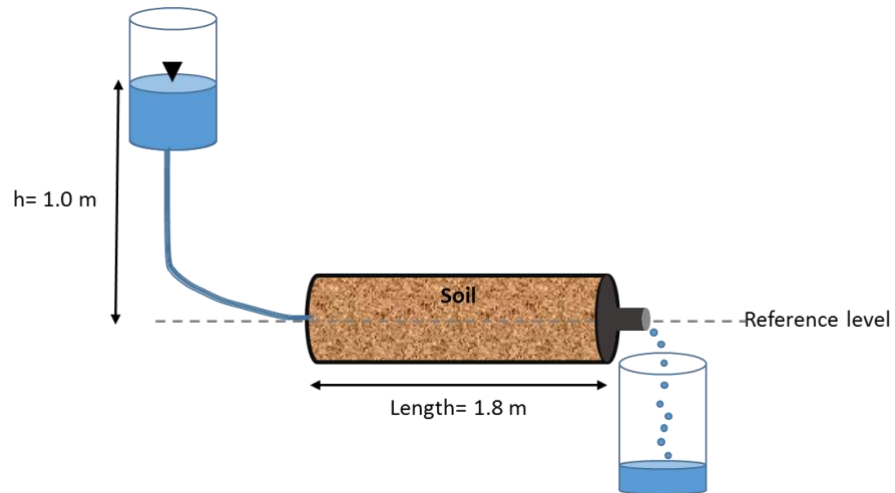


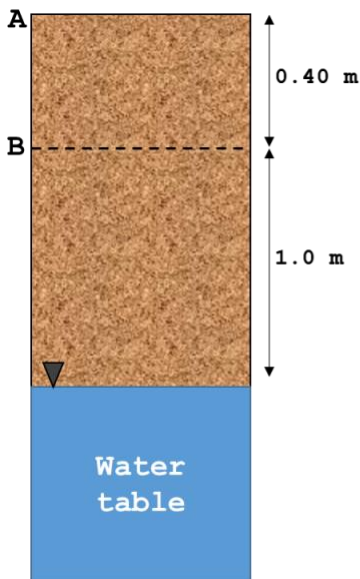
Image of two soils with different volumetric water content.

15. Calculate the flow through a soil macropore left by a plant root that is 5.0 mm in diameter and has a length of 10 cm. Assume a difference in hydraulic pressure of 2.0 cm.
16. Define hydraulic resistance. Write Darcy's Law in terms of hydraulic resistance for flow through a soil with two layers of differing hydraulic conductivity.
17. A steady water flux of 6.5 cm hr^{-1} is passing through the soil column below. What is the hydraulic conductivity of the soil?



Sketch showing the components for calculating the saturated hydraulic conductivity.

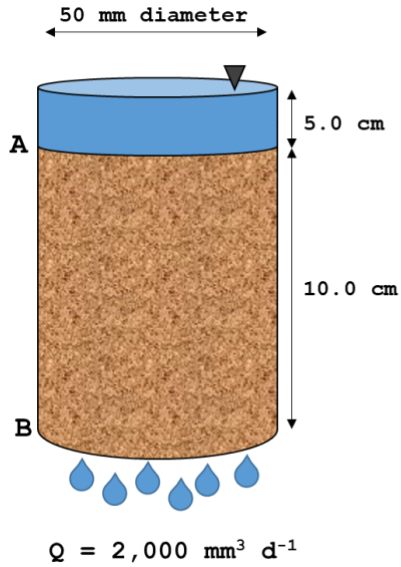
18. Look at the diagram below showing a deep, uniform sandy loam in equilibrium with a water table. Create a table showing the gravimetric, pressure, and total potentials at points A and B.



	A	B	Units
H_g			
H_p			
H_T			

Sketch showing a soil profile and the table of potentials.

19. A steady water flow is passing through the soil column below. What is the saturated hydraulic conductivity of the soil?



	A	B	Units
H_g			
H_p			
H_T			

20. A low-lying area in a cultivated field is often inundated with water after a heavy rain. To improve drainage, the farmer has installed a subsurface drainage system, which effectively maintains a free water surface at the 1.0 m soil depth. Estimate the soil water flux for the two following scenarios:

- A fully saturated soil, with 10.0 cm of ponded water on the surface, uniform silt loam soil from the surface to 1.0 m depth.
- A fully saturated soil, 10.0 cm of ponded water on the surface, silt loam from the surface to 0.60 m depth and clay loam from 0.60 to 1.0 m depth.

Note: Assume a K_{sat} value of 18.3 cm/day for the silt loam soil and 8.18 cm/day for the clay loam soil.